Duarte Lopes

Data scientist, machine learning and sports analytics

duarte.lopes99@hotmail.com duartelopes99.github.io linkedin.com/in/duarte-lopes-
github.com/DuarteLopes99 Portugal

Profile

Data scientist with a Bioengineering background and three years of industry experience building forecasting and applied machine learning systems, from data pipelines to production monitoring. Research roots in deep learning for computer vision at INESC TEC. Most at home on problems that run from raw data to something people use, especially in sport.

Experience

Data Scientist Aug 2023 to Aug 2026
Enlitia, Porto (remote)

- Designed, developed and deployed machine learning models for wind and solar power forecasting
- Built reusable ML frameworks for tabular energy data; analysed and validated data with SQL and Pandas
- Built Grafana dashboards to monitor model performance; applied NLP to text classification
- Migrated models and algorithms from local environments to Snowflake

Research Fellow, Deep Learning Feb 2023 to Aug 2023
INESC TEC, Porto (NewSpacePortugal project)

- Knowledge distillation for facial emotion recognition: compressed student at 91.3% on CK+, 2 points above training without distillation
- Benchmarked distillation, pruning and quantization; evaluated bias, fairness and overestimation

MSc thesis researcher Nov 2021 to Jul 2022
INESC TEC, Porto

Deep Neural Networks for Face-based Emotion Recognition (19/20): CNNs, transfer learning and Vision Transformers, with landmark-based face representations.

Research intern Nov 2020 to Jun 2021
INESC TEC, Porto

Presentation attack detection for face recognition, with explainable AI (19/20).

Education

Integrated MSc in Bioengineering, Biomedical Engineering 2017 to 2022
University of Porto, Faculty of Engineering (FEUP). Final grade 17/20.

Secondary education, Science and Technologies 2014 to 2017
Santa Maria da Feira

Publication

Lopes, D., Preto, I., Freire, D. Exploring Optimal Strategies for Small Hydro Power Forecasting: Training Periods and Methodological Variations. Engineering Proceedings 68(1), 27, 2024. doi.org/10.3390/engproc2024068027

Certificates

Time Series Forecasting with Python: Models to Production Nov 2026
Board Infinity, on Coursera

NLP: Natural Language Processing with Python Jun 2024
Udemy

Selected projects

- **Match video analysis:** single-camera football match to statistics, expected goals and highlight clips (detection, tracking, homography)
- **Season forecasts and GPS load:** Monte Carlo league forecasts and GPS training load analysis, with Streamlit dashboards
- **RAG architecture comparison:** four retrieval pipelines with ChromaDB and an LLM intent layer, plus an evaluation harness

Skills

Machine learning	PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, MediaPipe, NLP, time-series forecasting
Data	Python, SQL, Pandas, Snowflake, Grafana, Streamlit
Applied AI	Retrieval-augmented generation, sentence embeddings, ChromaDB, FastAPI
Also	HTML, CSS, PHP, Java, C, C++, C#, MATLAB, LabVIEW, LaTeX
Languages	Portuguese (native), English (C2)
Outside work	Federated football at regional level, trail and 10K road running, cycling, padel and tennis